CS1002 – Programming Fundamentals

Lecture # 03
Monday, August 29, 2022
FALL 2022
FAST – NUCES, Faisalabad Campus

Rizwan Ul Haq

Pseudo-Code: Decision Making

- **If-then**
- **General form:**

  if (condition is met) then

      statement(s)

Example:

  if temperature < 0 then

      wear a jacket

Algorithms and Pseudocode

- Take and input of four marks and calculate average. If the average is above 50 the student is declared pass.

Algorithms and Pseudocode

Algorithm:

Input a set of four marks

Calculate their average by summing and dividing the sum by 4

If average is above 50

    Print : "PASS"

Algorithms and Pseudocode

Pseudocode:

1.0 Declare M1, M2, M3, M4, average

2.0 Input M1, M2, M3, M4

3.0 average = (M1+ M2+ M3+ M4) / 4

4.0 if (average > 50) then

    4.1 Print "PASS"

5.0 endif

Pseudo-Code: Decision Making

If-then-else

General form:

```
if (condition is met) then
    statement(s)
else
    statements(s)
```

Example:

```
if (at work) then
    Dress formally
else
    Dress casually
```

Algorithms and Pseudocode

- Take and input of four marks and calculate average. If the average is above 50 the student is declared pass otherwise failed.

Algorithms and Pseudocode

Algorithm:

- Input a set of four marks
- Calculate their average by summing and dividing the sum by 4
- If average is above 50
    - Print : “PASS”

  else
    - Print “FAIL”

Algorithms and Pseudocode

Pseudocode:

1.0 Declare M1, M2, M3, M4, average

2.0 Input M1, M2, M3, M4

3.0 average = (M1+ M2+ M3+ M4) / 4

4.0 if (Grade > 50) then

    4.1 Print "PASS"

5.0 else

    5.1 Print "FAIL"

6.0 endif

Pseudocode – Comparing two numbers

```
1.0 start
2.0 declare n1, n2
3.0 input n1,n2
4.0 if (n1 > n2)
        4.1 print "n1 is greater"
5.0 endif
6.0 if (n2 > n1)
        6.1 print "n2 is greater"
7.0 if (n2 = n1)
        7.1 print "they are equal"
8.0 endif
9.0 end
```

Pseudo-Code: Fast Food Example

- Use pseudo-code to specify the algorithm for a person who is ordering food at a fast food restaurant

- At the food counter, the person can either order or not order the following items:
- a burger
- Fries
- a drink
- After placing her order the person then goes to the cashier and pays the bill

Algorithm: Fast Food Example

1.0 Approach counter

2.0 if want burger then

    2.1. order burger

3.0 if want fries then

    3.1. order fries

4.0 if want drink then

    4.1 order drink

5.0 Pay cashier

Pseudo-Code: Fast Food Example (Computer)

1.0 Declare order_burger, order_fries, order_drink, Bill = 0

2.0 Print ‘Do you want to order burger?’

3.0 Input order_burger

4.0 if order_burger = yes then

    4.1. Bill ← Bill + 300

5.0 endif

6.0 Print ‘Do you want to order fries?’

7.0 Input order_fries

8.0 if order_fries = yes then

    8.1. Bill ← Bill + 100

9.0 Output 'Do you want to order drink?’

10.0 Input order_drink

11.0 If order_drink = yes then

    11.1. Bill ← Bill + 50

12.0 Pay Bill to cashier

Questions